

CHAPPIDI RAMA PRAKASH

Full Stack Developer | Backend Engineer | Cloud & Real-Time Systems

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Portfolio: portfolio-eight-vidid-6hjhyksnoy.vercel.app

PROFESSIONAL SUMMARY

Results-driven Full Stack / Backend Developer with hands-on experience designing and deploying scalable, production-grade systems. Proficient in building RESTful APIs, real-time communication platforms, and cloud-hosted services using Node.js, Express.js, PostgreSQL, Redis, Socket.IO, and AWS. Demonstrated ability to architect end-to-end solutions — from database schema design to EC2 deployment — for service booking, live tracking, and subscription-based products. Solved 200+ LeetCode problems (Easy, Medium & Hard) with strong focus on Graphs and Linked Lists. Passionate about clean architecture, performance optimization, and delivering reliable systems at scale.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), Python

Backend: Node.js, Express.js, REST APIs, MVC Architecture, RBAC, JWT Authentication, Middleware Design

Real-Time: Socket.IO, Live GPS Tracking, Bidirectional Communication, Event-Driven Architecture

Databases: PostgreSQL, Redis — Schema Design, Query Optimization, Indexing, Caching, Session Management, Stored Procedures, Triggers

Cloud & DevOps: AWS EC2 (instance setup, SSH, security groups), AWS S3 (media storage, presigned URLs), Nginx (reverse proxy, SSL), PM2

Data & Visualization: Python, Pandas, Matplotlib, Seaborn, Web Scraping (Requests, BeautifulSoup)

DSA: 200+ LeetCode problems (Easy, Medium & Hard) — Graphs, Linked Lists, Trees, Arrays, DP, Sliding Window, Backtracking

Tools: Git, GitHub, Postman, Linux CLI, VS Code

Integrations: Payment Gateway APIs, Third-Party Service APIs, Push Notifications, Webhook Handlers

PROFESSIONAL EXPERIENCE

Database Intern • Algonox Pvt. Ltd.

Oct 2024 – Mar 2025 | Database Management, Migration & Backend Integration

- Executed a full-scale data migration from Microsoft SQL Server (MSSQL) to PostgreSQL, covering schema translation, data type mapping, and post-migration validation — ensuring zero data loss and consistent query performance.
- Designed and authored SQL scripts, stored procedures, and triggers to automate repetitive data operations, reducing manual overhead for backend teams.
- Participated in schema normalization and relational database design applying 3NF principles to minimize redundancy and support scalable backend services.
- Collaborated with backend developers to integrate optimized database layers with RESTful API endpoints, improving response times and maintainability.
- Conducted performance profiling of slow-running queries using EXPLAIN ANALYZE and applied indexing strategies to improve query execution plans.

Freelance Full Stack Developer • CoreFix MEP Services

Sep 2025 – Jan 2026 | Node.js, Express.js, PostgreSQL, AWS EC2, AWS S3, JWT, REST APIs

- Architected and developed a multi-module backend system for a Mechanical, Electrical & Plumbing (MEP) service company — handling service booking, AMC subscription management, payment workflows, user roles, and technician job assignment.
- Designed normalized PostgreSQL schemas supporting complex relationships between customers, service requests, technician profiles, job statuses, and billing records.
- Built secure REST API endpoints with JWT-based authentication and RBAC separating Admin, Technician, and Customer permissions across all routes.
- Deployed on AWS EC2: configured Ubuntu server, Nginx reverse proxy, SSL, and PM2 process management for zero-downtime restarts.

- Integrated AWS S3 for scalable media storage with pre-signed URLs for secure, time-limited upload and retrieval of service reports and documents.
- Implemented payment workflow APIs for service billing, subscription renewal reminders, and payment status tracking with third-party provider integration.

Freelance Full Stack Developer • Vegaa – Real-Time Service Booking Platform

Feb 2026 – Apr 2026 | Node.js, Express.js, Socket.IO, Redis, PostgreSQL, AWS EC2, Nginx

- Designed and built a real-time on-demand service booking platform (Uber/Rapido-inspired) connecting users with nearby providers with live status updates and GPS tracking throughout the session lifecycle.
- Implemented bidirectional event-driven communication using Socket.IO — handling booking requests, provider acceptance/rejection, state transitions (requested → accepted → en-route → completed), and in-app messaging.
- Built live GPS location tracking with continuous coordinate emission from provider to user, enabling real-time map updates and ETA estimation.
- Leveraged Redis for session management, socket state persistence, and pub/sub horizontal scaling — ensuring correct event routing across multiple server instances.
- Engineered provider matching logic using optimized PostgreSQL spatial queries to rank nearby providers by geolocation, service type, and availability.
- Deployed on AWS EC2 behind an Nginx reverse proxy with WebSocket upgrade support for persistent socket connections in production.

PROJECTS

CoreFix MEP Services – Service Management Platform

Sep 2025 – Jan 2026 | Node.js, Express.js, PostgreSQL, AWS EC2, AWS S3, JWT, REST APIs

- End-to-end backend platform for an MEP company covering service booking, AMC subscriptions, technician dispatch, and payment workflows.
- RBAC with JWT separating Admin, Technician, and Customer roles with secure route-level access control.
- Deployed on AWS EC2 with Nginx, SSL, PM2, and AWS S3 pre-signed URL integration for media storage.

Vegaa – Real-Time On-Demand Service Booking Platform

Feb 2026 – Apr 2026 | Node.js, Express.js, Socket.IO, Redis, PostgreSQL, AWS EC2, Nginx

- Real-time booking platform with live GPS tracking, Socket.IO event-driven communication, and full session lifecycle management.
- Redis pub/sub for horizontal socket scaling; PostgreSQL spatial queries for proximity-based provider matching.
- Production-deployed on AWS EC2 with Nginx configured for WebSocket upgrade support.

YouTube Channel Analysis – Web Scraping & Data Visualization

2025 | Python, Requests, BeautifulSoup, Pandas, Matplotlib, Seaborn

- Built a web scraping pipeline in Python to extract YouTube channel metrics — video titles, views, likes, upload frequency, and engagement rates — for comprehensive channel performance analysis.
- Processed and cleaned raw scraped data using Pandas, handling missing values, normalizing formats, and structuring datasets for downstream analysis.
- Visualized insights using Matplotlib and Seaborn — including view trend charts, engagement rate distributions, upload frequency heatmaps, and top-performing content breakdowns.
- Derived actionable insights on content performance patterns, peak upload times, and audience engagement trends through exploratory data analysis (EDA).

DSA & COMPETITIVE PROGRAMMING

LeetCode Problem Solving

Ongoing | 200+ Problems Solved — Easy, Medium & Hard

- Solved 200+ problems on LeetCode across Easy, Medium, and Hard difficulty levels with consistent focus on core data structures and algorithmic patterns.
- Strong proficiency in Graphs (BFS, DFS, shortest path, cycle detection, topological sort) and Linked Lists (reversal, cycle detection, merge, two-pointer techniques).
- Practiced Trees (BST, traversals), Arrays, Sliding Window, Two Pointers, Recursion, Backtracking, and Dynamic Programming.
- Emphasis on writing optimized, clean solutions with analysis of time and space complexity for each approach.

EDUCATION

B.Tech – Computer Science & Engineering • RGUKT IIIT Srikakulam

June 2021 – May 2025 | CGPA: 8.2 / 10.0

- Rajiv Gandhi University of Knowledge Technologies
- Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Software Engineering, OOP

Pre-University Course (PUC / Intermediate) • Board of Intermediate Education, AP

Aug 2019 – Apr 2021 | CGPA: 8.9 / 10.0

CERTIFICATIONS

- SQL Certification — Great Learning | 2024
- MongoDB Certification — MongoDB University (Official) | 2023